BPS-17 Lecturer Mathematics
AJKPSC / FPSC / PPSC / KPSC
Comprehensive Study Plan

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Linear Algebra & Vector Spaces

Topics Covered:

- Matrices & Determinants
- Vector Spaces & Subspaces
- Basis & Dimension
- Rank–Nullity Theorem
- Kernel & Range
- Eigenvalues & Eigenvectors
- Inner Product Spaces
- Orthogonality & Projections
- Systems of Linear Equations
- Linear Independence
- Row, Column & Null Space
- Linear Transformations
- Matrix Representations
- Diagonalization
- Gram–Schmidt Process
- Quadratic Forms

Each section: **Concept Summary → Key Formulas → MCQs with Full Explanations**

Reference Material — PSC 2021–26

Contents

1. Matrices, Determinants & Systems of Equations

A **matrix** $A \in M_{m\times n}(\mathbb{F})$ is a rectangular array of scalars. Key operations include addition, scalar multiplication, and matrix multiplication. The **determinant** $\det(A)$ is defined for square matrices and measures invertibility. A matrix A is **invertible** (non-singular) if and only if $\det(A) \neq 0$.

Key Formulas & Results

- $\det(AB) = \det(A)\det(B)$
- $\det(A^{-1}) = \dfrac{1}{\det(A)}$
- $\det(A^T) = \det(A)$
- $\det(kA) = k^n \det(A) \quad (A \text{ is } n \times n)$
- **Cramer's Rule:** $x_i = \dfrac{\det(A_i)}{\det(A)}$ where A_i replaces column i by $\mathbf{b}$
- $A^{-1} = \dfrac{1}{\det(A)} \operatorname{adj}(A)$

1.1. MCQs

Q1. If A is a 3×3 matrix with $\det(A) = 5$, then $\det(3A)$ equals:

(A) 15

(B) 45

(C) 135

(D) 25

Correct Answer: (C) 135

For an $n \times n$ matrix, $\det(kA) = k^n \det(A)$. Here $n = 3$ and $k = 3$:

$$\det(3A) = 3^3 \cdot \det(A) = 27 \times 5 = 135.$$

Q2. Which of the following is *not* a property of the determinant?

(A) $\det(AB) = \det(A)\det(B)$

(B) $\det(A + B) = \det(A) + \det(B)$

(C) $\det(A^T) = \det(A)$

(D) $\det(A^{-1}) = [\det(A)]^{-1}$

Correct Answer: (B)

The determinant is *not* linear over matrix addition. A simple counterexample:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \det(A+B) = 1 \neq \det(A) + \det(B) = 0.$$

Options (A), (C), (D) are all standard determinant identities.

Q3. The system $AX = B$ has a unique solution if and only if:

(A) $\det(A) = 0$

(B) $\det(A) \neq 0$

(C) $B = \mathbf{0}$

(D) A is symmetric

Correct Answer: (B)

A square system $AX = B$ has a unique solution $\iff$ A is invertible $\iff$ $\det(A) \neq 0$. When $\det(A) = 0$, the system either has no solution or infinitely many solutions.

Q4. If A is an invertible $n \times n$ matrix, then $\operatorname{rank}(A)$ equals:

(A) 0

(B) $n - 1$

(C) n

(D) Depends on entries of A

Correct Answer: (C) n

A invertible $\Rightarrow$ all rows (and columns) are linearly independent $\Rightarrow$ $\operatorname{rank}(A) = n$ (full rank).

Q5. The matrix $\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$ is:

(A) Invertible with $\det = 4$

(B) Invertible with $\det = 8$

(C) Singular (non-invertible)

(D) Orthogonal

Correct Answer: (C) Singular

$\det \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix} = (2)(2) - (4)(1) = 4 - 4 = 0$. Since $\det = 0$, the matrix is **singular** (not invertible). Notice row 2 is exactly $\frac{1}{2}$ row 1, so the rows are linearly dependent.

When checking invertibility quickly: look for proportional rows or columns — if any two rows/columns are proportional, $\det = 0$ immediately.

2. Vector Spaces & Subspaces

A **vector space** V over a field $\mathbb{F}$ is a set with two operations (vector addition and scalar multiplication) satisfying 10 axioms: commutativity, associativity, identity, inverses, distributivity, etc. A **subspace** $W \subseteq V$ is a subset closed under addition and scalar multiplication (equivalently: W is non-empty, and $\alpha\mathbf{u} + \beta\mathbf{v} \in W$ for all $\mathbf{u}, \mathbf{v} \in W$, $\alpha, \beta \in \mathbb{F}$).

Key Formulas & Results

Subspace Test (Three Conditions):

1. $\mathbf{0} \in W$ (contains zero vector)
2. $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$ (closed under addition)
3. $\mathbf{u} \in W$, $c \in \mathbb{F} \Rightarrow c\mathbf{u} \in W$ (closed under scalar multiplication)

Note: Conditions 2 & 3 combine to: $\alpha\mathbf{u} + \beta\mathbf{v} \in W$.

2.1. MCQs

Q6. Which of the following is *not* a vector space over $\mathbb{R}$?

(A) $\mathbb{R}^n$ with standard operations

(B) The set of all $m \times n$ real matrices

(C) The set of all polynomials $p(x)$ with $p(0) = 1$

(D) The set of all continuous functions on $[0, 1]$

Correct Answer: (C)

The set $\{p(x) : p(0) = 1\}$ is *not* a vector space because it does not contain the zero polynomial ($0(x) = 0$, so $0(0) = 0 \neq 1$). It also fails closure: if $p(0) = 1$ and

$q(0) = 1$, then $(p + q)(0) = 2 \neq 1$.

Q7. Which subset of $\mathbb{R}^2$ is a subspace?

(A) $W = \{(x, y) : x + y = 1\}$

(B) $W = \{(x, y) : xy = 0\}$

(C) $W = \{(x, y) : x = 2y\}$

(D) $W = \{(x, y) : x^2 + y^2 = 1\}$

Correct Answer: (C) $W = \{(x, y) : x = 2y\}$

- (A): $(0, 0)$ gives $0 + 0 = 0 \neq 1$ — fails zero-vector test.
- (B): $(1, 0), (0, 1) \in W$ but $(1, 1) \notin W$ — not closed under addition.
- (C): Contains $\mathbf{0}$ ($x = 2y = 0$); if $x_1 = 2y_1$ and $x_2 = 2y_2$ then $(x_1 + x_2) = 2(y_1 + y_2)$; $cx = 2(cy)$. **Subspace!**
- (D): A circle — doesn't contain $\mathbf{0}$.

Q8. Let $W = \{(a, b, c) \in \mathbb{R}^3 : a + b + c = 0\}$. The dimension of W is:

(A) 0

(B) 1

(C) 2

(D) 3

Correct Answer: (C) 2

W is the solution set of one homogeneous equation in 3 unknowns. The constraint $c = -a - b$ gives free variables a, b, so $\dim W = 2$. A basis is $\{(1, 0, -1), (0, 1, -1)\}$.

Q9. The intersection $W_1 \cap W_2$ of two subspaces W_1, W_2 of V is:

(A) Always equal to V

(B) Always a subspace of V

(C) A subspace only if $W_1 \subseteq W_2$

(D) Never a subspace

Correct Answer: (B)

$W_1 \cap W_2$ always contains $\mathbf{0}$, is closed under addition (if $\mathbf{u}, \mathbf{v}$ are in both, so is $\mathbf{u} + \mathbf{v}$), and is closed under scalar multiplication. Hence it is always a subspace. Note: the *union* $W_1 \cup W_2$ is generally *not* a subspace.

3. Linear Independence, Basis & Dimension

Vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ are **linearly independent** if $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$ implies $c_1 = \cdots = c_k = 0$. A **basis** for V is a linearly independent spanning set. All bases of V have the same cardinality, called the **dimension** $\dim V$.

Key Formulas & Results

- $\dim(\mathbb{R}^n) = n; \quad \dim(M_{m\times n}) = mn; \quad \dim(P_n) = n+1$
- Any $n+1$ vectors in an n-dim space are linearly dependent
- Any n linearly independent vectors in an n-dim space form a basis
- **Rank–Nullity:** $\text{rank}(A) + \text{null}(A) = n \quad (A \text{ is } m \times n)$

3.1. MCQs

Q10. The vectors $(1,2,3)$, $(4,5,6)$, $(7,8,9)$ in $\mathbb{R}^3$ are:

(A) Linearly independent and form a basis for $\mathbb{R}^3$

(B) Linearly dependent

(C) Orthogonal

(D) Forming a basis for a 2-dimensional subspace

Correct Answer: (B) Linearly dependent

Notice $(7,8,9) = 2(4,5,6) - (1,2,3)$, so the three vectors are linearly dependent. Equivalently, the 3×3 matrix formed by these rows has $\det = 0$.

Q11. The dimension of the vector space P_3 (polynomials of degree ≤ 3) is:

(A) 3

(B) 4

(C) ∞

(D) 2

Correct Answer: (B) 4

The standard basis for P_3 is $\{1, x, x^2, x^3\}$, which has 4 elements. In general, $\dim P_n = n + 1$.

Q12. If $S = \{v_1, v_2, v_3\}$ is a basis for V and $w = 3v_1 - 2v_2 + v_3$, the coordinate vector of w relative to S is:

(A) $(1, -2, 3)$

(B) $(3, -2, 1)$

(C) $(-2, 3, 1)$

(D) $(3, 2, -1)$

Correct Answer: (B) $(3, -2, 1)$

The coordinate vector lists the coefficients in the order of the basis elements: $w = 3v_1 + (-2)v_2 + 1 \cdot v_3$, so $[w]_S = (3, -2, 1)$.

Q13. A set of 5 vectors in $\mathbb{R}^4$ must be:

(A) A basis for $\mathbb{R}^4$

(B) Linearly independent

(C) Linearly dependent

(D) A spanning set for $\mathbb{R}^4$

Correct Answer: (C) Linearly dependent

Any set of more than $\dim V$ vectors in V must be linearly dependent. Since $\dim \mathbb{R}^4 = 4$, any 5 vectors in $\mathbb{R}^4$ are automatically linearly dependent.

Q14. If A is a 4×6 matrix with $\text{rank}(A) = 3$, then the nullity of A is:

(A) 1

(B) 2

(C) 3

(D) 4

Correct Answer: (C) 3

By the **Rank–Nullity Theorem**: $\text{rank}(A) + \text{null}(A) = n$ where n is the number of columns.

$$\text{null}(A) = 6 - \text{rank}(A) = 6 - 3 = 3.$$

Remember: Rank–Nullity uses the number of **columns** (n), NOT the number of rows.

4. Linear Transformations

A map $T : V \to W$ is a **linear transformation** if:

$$T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}) \quad \forall\, \mathbf{u}, \mathbf{v} \in V,\ \alpha, \beta \in \mathbb{F}.$$

The **kernel** (null space) $\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}$ and the **range** (image) $\operatorname{Im}(T) = \{T(\mathbf{v}) : \mathbf{v} \in V\}$ are subspaces. T is **injective** (one-to-one) $\iff \ker(T) = \{\mathbf{0}\}$; T is **surjective** (onto) $\iff \operatorname{Im}(T) = W$.

Key Formulas & Results

- $\dim(\ker T) + \dim(\operatorname{Im} T) = \dim V$ (Dimension Theorem)
- T is bijective $\iff$ T is an **isomorphism**
- $[T(\mathbf{v})]_C = [T]_{BC}\,[\mathbf{v}]_B$ (matrix representation)
- $T(\mathbf{0}) = \mathbf{0}$ always for any linear transformation

4.1. MCQs

Q15. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(x, y) = (x + y,\ x - y)$. Which is true?

(A) T is not linear

(B) $\ker(T) = \mathbb{R}^2$

(C) T is linear and bijective

(D) $\operatorname{Im}(T) = \{(0, 0)\}$

Correct Answer: (C) T is linear and bijective

Linearity: $T(\alpha(x_1, y_1) + \beta(x_2, y_2)) = \alpha T(x_1, y_1) + \beta T(x_2, y_2)$ — holds by inspection.
Matrix: $[T] = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $\det = -2 \neq 0$, so T is bijective.

Q16. If $T : \mathbb{R}^3 \to \mathbb{R}^2$ is linear, then T cannot be:

(A) Onto (surjective)

(B) One-to-one (injective)

(C) The zero map

(D) Represented by a matrix

Correct Answer: (B) One-to-one (injective)

By the Dimension Theorem: $\dim(\ker T) = 3 - \dim(\operatorname{Im} T) \geq 3 - 2 = 1 > 0$. So

$\ker(T) \neq \{\mathbf{0}\}$, meaning T cannot be injective. It can, however, be surjective (onto $\mathbb{R}^2$).

Q17. The kernel of $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(x, y, z) = (x + y,\ y + z,\ 0)$ has dimension:

(A) 0

(B) 1

(C) 2

(D) 3

Correct Answer: (B) 1

$\ker(T)$: solve $x + y = 0$, $y + z = 0 \Rightarrow x = -y$, $z = -y$. Free variable: y. So $\ker(T) = \operatorname{span}\{(-1, 1, -1)\}$ and $\dim(\ker T) = 1$. (Check: $\dim(\operatorname{Im} T) = 2$, $1 + 2 = 3 = \dim \mathbb{R}^3$)

Q18. Which of the following maps $T : \mathbb{R}^2 \to \mathbb{R}^2$ is *not* linear?

(A) $T(x, y) = (2x - y,\ 3x + 4y)$

(B) $T(x, y) = (x, 0)$

(C) $T(x, y) = (x + 1,\ y)$

(D) $T(x, y) = (y, x)$

Correct Answer: (C) $T(x, y) = (x + 1, y)$

A linear map must satisfy $T(\mathbf{0}) = \mathbf{0}$. Here $T(0, 0) = (1, 0) \neq (0, 0)$. Hence T is **not** linear (it is an affine map, not a linear one).

5. Eigenvalues & Eigenvectors

A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of $A \in M_{n\times n}(\mathbb{F})$ if there exists a non-zero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$. Such $\mathbf{x}$ is called an **eigenvector**. Eigenvalues are found from the **characteristic equation**:

$$\det(A - \lambda I) = 0.$$

The eigenspace corresponding to λ is $E_\lambda = \ker(A - \lambda I)$.

Key Formulas & Results

- Characteristic polynomial: $p(\lambda) = \det(A - \lambda I)$

- $\sum_i \lambda_i = \text{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$
- $\prod_i \lambda_i = \det(A)$
- A is diagonalizable $\iff$ A has n linearly independent eigenvectors
- Symmetric matrices are always diagonalizable over $\mathbb{R}$ (Spectral Theorem)
- $A = PDP^{-1}$ where columns of P are eigenvectors, D is diagonal

5.1. MCQs

Q19. The eigenvalues of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ are:

(A) 4 and 3

(B) 5 and 2

(C) 6 and 1

(D) 7 and 0

Correct Answer: (B) 5 **and** 2

$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - (1)(2) = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0.$
Eigenvalues: $\lambda_1 = 5,\ \lambda_2 = 2.$
Check: $\text{tr}(A) = 4 + 3 = 7 = 5 + 2$, $\det(A) = 12 - 2 = 10 = 5 \times 2$.

Q20. If $\lambda = 0$ is an eigenvalue of A, then:

(A) A is invertible

(B) A is the identity matrix

(C) A is singular

(D) $\text{tr}(A) = 0$

Correct Answer: (C) A **is singular**

$\lambda = 0$ is an eigenvalue $\Rightarrow \det(A - 0 \cdot I) = \det(A) = 0 \Rightarrow A$ is singular (not invertible).

Q21. The matrix $A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$ has eigenvectors:

(A) Only $(1, 0)$ and $(0, 1)$

(B) Every non-zero vector in $\mathbb{R}^2$

(C) Only $(1, 1)$

(D) No real eigenvectors

Correct Answer: (B) Every non-zero vector in $\mathbb{R}^2$

$A = 3I$, so $A\mathbf{x} = 3\mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^2$. Every non-zero vector is an eigenvector with eigenvalue $\lambda = 3$. The eigenspace is all of $\mathbb{R}^2$.

Q22. If A has eigenvalue λ with eigenvector $\mathbf{x}$, then A^3 has eigenvalue:

(A) 3λ

(B) λ^3

(C) $3\lambda^2$

(D) $\lambda + 3$

Correct Answer: (B) λ^3

$A\mathbf{x} = \lambda\mathbf{x} \Rightarrow A^2\mathbf{x} = \lambda A\mathbf{x} = \lambda^2\mathbf{x} \Rightarrow A^3\mathbf{x} = \lambda^3\mathbf{x}$. In general, $A^k\mathbf{x} = \lambda^k\mathbf{x}$.

Q23. A 3×3 real symmetric matrix has:

(A) At most 1 real eigenvalue

(B) Exactly 3 real eigenvalues (counting multiplicity)

(C) Only complex eigenvalues

(D) No real eigenvectors

Correct Answer: (B) Exactly 3 real eigenvalues (counting multiplicity)

By the **Spectral Theorem**, every real symmetric matrix has all eigenvalues real. Since the characteristic polynomial has degree 3, there are exactly 3 real eigenvalues (counting algebraic multiplicity).

For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$: eigenvalues satisfy $\lambda^2 - (a+d)\lambda + (ad - bc) = 0$, i.e., $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$. Memorize this!

6. Diagonalization

A is **diagonalizable** if $A = PDP^{-1}$ for some invertible P and diagonal D. This happens $\iff$ A has n linearly independent eigenvectors. If eigenvalues are all distinct, A is automatically diagonalizable. The diagonal entries of D are the eigenvalues; columns of P are corresponding eigenvectors.

6.1. MCQs

Q24. Which matrix is diagonalizable?

(A) $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

(B) $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$

(C) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (over $\mathbb{R}$)

(D) $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

Correct Answer: (B)

(B) is already diagonal — it is diagonalizable with eigenvalues 2 and 3. (A) is a Jordan block with repeated eigenvalue $\lambda = 1$ but only one independent eigenvector — *not* diagonalizable over $\mathbb{R}$ (or $\mathbb{C}$). (C) has characteristic polynomial $\lambda^2 + 1 = 0$, no real eigenvalues — *not* diagonalizable over $\mathbb{R}$.

Q25. If $A = PDP^{-1}$, then A^{10} equals:

(A) $P^{10}D^{10}P^{-10}$

(B) $PD^{10}P^{-1}$

(C) $10\,PDP^{-1}$

(D) D^{10}

Correct Answer: (B) $PD^{10}P^{-1}$

$A^2 = (PDP^{-1})(PDP^{-1}) = PD(P^{-1}P)DP^{-1} = PD^2P^{-1}$.
By induction, $A^k = PD^kP^{-1}$. Since D is diagonal, D^{10} is trivial to compute (raise each diagonal entry to the 10th power).

7. Inner Product Spaces & Orthogonality

An **inner product** on V over $\mathbb{R}$ is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying: symmetry, linearity in the first argument, and positive definiteness. The **norm** is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$. Vectors are **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. An **orthonormal basis** is a basis of unit mutually orthogonal vectors.

Key Formulas & Results

- **Cauchy–Schwarz:** $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$
- **Projection:** $\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \dfrac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v}$
- **Gram–Schmidt:** $\mathbf{e}_k = \mathbf{v}_k - \displaystyle\sum_{j<k} \operatorname{proj}_{\mathbf{e}_j} \mathbf{v}_k$, then normalize
- **Pythagorean:** $\mathbf{u} \perp \mathbf{v} \Rightarrow \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$

7.1. MCQs

Q26. In $\mathbb{R}^3$ with the standard inner product, the projection of $\mathbf{u} = (1, 2, 3)$ onto $\mathbf{v} = (1, 1, 1)$ is:

(A) $(1, 2, 3)$

(B) $(2, 2, 2)$

(C) $(1, 1, 1)$

(D) $(3, 3, 3)$

Correct Answer: (B) $(2, 2, 2)$

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v} = \frac{1+2+3}{1+1+1}(1, 1, 1) = \frac{6}{3}(1, 1, 1) = 2(1, 1, 1) = (2, 2, 2).$$

Q27. Vectors $\mathbf{u} = (1, -1, 0)$ and $\mathbf{v} = (1, 1, 0)$ in $\mathbb{R}^3$ are:

(A) Parallel

(B) Orthogonal

(C) Equal in norm

(D) Both (B) and (C)

Correct Answer: (D) Both (B) and (C)

$\langle \mathbf{u}, \mathbf{v} \rangle = 1 \cdot 1 + (-1) \cdot 1 + 0 \cdot 0 = 0$ — orthogonal .
$\|\mathbf{u}\| = \sqrt{2},\ \|\mathbf{v}\| = \sqrt{2}$ — equal norms .

Q28. For any vectors $\mathbf{u}, \mathbf{v}$ in an inner product space, the Cauchy–Schwarz inequality states:

(A) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

(B) $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$

(C) $\langle \mathbf{u}, \mathbf{v} \rangle \geq 0$

(D) $\|\mathbf{u} - \mathbf{v}\| \leq \|\mathbf{u}\| - \|\mathbf{v}\|$

Correct Answer: (B)

The **Cauchy–Schwarz Inequality** is $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$, with equality iff $\mathbf{u}$ and $\mathbf{v}$ are scalar multiples. Option (A) is the Triangle Inequality (also true, but that's a different result).

Q29. The Gram–Schmidt process applied to a linearly independent set produces:

(A) An orthogonal basis with the same span

(B) A basis with all equal norms

(C) An orthogonal basis for all of V

(D) A set with fewer vectors

Correct Answer: (A)

Gram–Schmidt converts a linearly independent set $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ into an **orthogonal** (or orthonormal, after normalizing) set $\{\mathbf{e}_1, \ldots, \mathbf{e}_k\}$ with $\operatorname{span}\{\mathbf{e}_1, \ldots, \mathbf{e}_k\} = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$. It does *not* change the number of vectors.

8. Row Space, Column Space & Null Space

If $T : V \to W$ is a linear transformation (or A is $m \times n$), then:

$$\dim(\ker T) + \dim(\operatorname{Im} T) = \dim V \quad \Longleftrightarrow \quad \operatorname{null}(A) + \operatorname{rank}(A) = n.$$

8.1. MCQs

Q30. For $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, the rank of A is:

(A) 3

(B) 1

(C) 2

(D) 0

Correct Answer: (C) 2

Row reduce: $R_2 \leftarrow R_2 - 4R_1$, $R_3 \leftarrow R_3 - 7R_1$:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix}.$$

Two non-zero rows $\Rightarrow \text{rank}(A) = 2$.

Q31. The column space of A and the row space of A always have the same:

(A) Set of vectors

(B) Dimension

(C) Span in $\mathbb{R}^n$

(D) Basis vectors

Correct Answer: (B) Dimension

By the definition of rank, $\dim(\text{Col}\, A) = \dim(\text{Row}\, A) = \text{rank}(A)$. The actual *sets* are different (row space $\subseteq \mathbb{R}^n$, column space $\subseteq \mathbb{R}^m$), but they share the same dimension.

Q32. If A is a 5×7 matrix with $\text{rank}(A) = 4$, then the dimension of the null space of A^T is:

(A) 1

(B) 3

(C) 4

(D) 2

Correct Answer: (A) 1

A^T is a 7×5 matrix. $\text{rank}(A^T) = \text{rank}(A) = 4$. By Rank–Nullity for A^T: $\text{null}(A^T) =$

$5 - \text{rank}(A^T) = 5 - 4 = 1$. (The null space of A^T is called the **left null space** of A.)

9. Quadratic Forms & Symmetric Matrices

A **quadratic form** on $\mathbb{R}^n$ is $Q(\mathbf{x}) = \mathbf{x}^T A\mathbf{x}$ where A is symmetric. It is **positive definite** if $Q(\mathbf{x}) > 0$ for all $\mathbf{x} \neq \mathbf{0}$; equivalently, all eigenvalues of A are positive.

Key Formulas & Results

- Positive definite $\iff$ all eigenvalues > 0 $\iff$ all leading principal minors > 0
- Negative definite $\iff$ all eigenvalues < 0
- Indefinite $\iff$ eigenvalues of both signs
- Real symmetric $\Rightarrow$ orthogonally diagonalizable: $A = Q\Lambda Q^T$, Q orthogonal

9.1. MCQs

Q33. The quadratic form $Q(x, y) = 2x^2 + 3y^2$ is:

(A) Negative definite

(B) Indefinite

(C) Positive definite

(D) Positive semi-definite only

Correct Answer: (C) Positive definite

$Q(x, y) = 2x^2 + 3y^2 \geq 0$ for all (x, y), and $Q(x, y) = 0 \iff x = y = 0$. The matrix is $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ with eigenvalues $2 > 0$ and $3 > 0$ — positive definite.

Q34. A real symmetric matrix A satisfies $A = A^T$. Its eigenvectors corresponding to *distinct* eigenvalues are:

(A) Parallel

(B) Orthogonal

(C) Linearly dependent

(D) Equal in magnitude

Correct Answer: (B) Orthogonal

This is a key result of the **Spectral Theorem**: for a real symmetric matrix, eigenvectors corresponding to *distinct* eigenvalues are always **orthogonal**. Proof sketch: if $A\mathbf{x} = \lambda\mathbf{x}$ and $A\mathbf{y} = \mu\mathbf{y}$ with $\lambda \neq \mu$, then $\lambda\langle\mathbf{x}, \mathbf{y}\rangle = \langle A\mathbf{x}, \mathbf{y}\rangle = \langle\mathbf{x}, A\mathbf{y}\rangle = \mu\langle\mathbf{x}, \mathbf{y}\rangle$, so $(\lambda - \mu)\langle\mathbf{x}, \mathbf{y}\rangle = 0$, giving $\langle\mathbf{x}, \mathbf{y}\rangle = 0$.

10. Quick-Reference Summary Table

Concept	Key Fact
Subspace test	Closed under + and scalar mult.; contains $\mathbf{0}$
Basis	Linearly independent spanning set
Dimension	Number of elements in any basis
Rank–Nullity	$\text{rank}(A) + \text{null}(A) = n$ (columns of A)
Eigenvalue	$\det(A - \lambda I) = 0$; trace $= \sum \lambda_i$; det $= \prod \lambda_i$
Diagonalizable	$\iff$ n linearly independent eigenvectors
Cauchy–Schwarz	$\lvert\langle\mathbf{u}, \mathbf{v}\rangle\rvert \leq \lVert\mathbf{u}\rVert\,\lVert\mathbf{v}\rVert$
Positive Definite	All eigenvalues > 0
Spectral Theorem	Real symmetric $\Rightarrow$ all eigenvalues real, orthog. diag.
Gram–Schmidt	Converts LI set to orthogonal set with same span

Exam Practice Resource

For further exam practice, the following resource provides a comprehensive collection of solved MCQs:
Access the material here

Best of Luck in Your PSC Examination!

Practice these MCQs thoroughly. Focus on Rank–Nullity, Eigenvalues, and the Spectral Theorem — they appear most frequently.